Available online at www.sciencedirect.com
SCIENCE DIRECT®
Nuclear Engineering and Design 224 (2003) 149–160
Nuclear Engineering and Design
www.elsevier.com/locate/nucengdes

Thermal-hydraulic analyses of transients in an actinide-burner reactor cooled by forced convection of lead–bismuth

Cliff B. Davis*

Idaho National Engineering and Environmental Laboratory, P.O. Box 1625, Idaho Falls, ID 83415-3890, USA

Received 14 November 2002; received in revised form 21 February 2003; accepted 1 April 2003

Abstract

The Idaho National Engineering and Environmental Laboratory (INEEL) and the Massachusetts Institute of Technology (MIT) are investigating the suitability of lead or lead–bismuth cooled fast reactors for producing low-cost electricity as well as for actinide burning. The current analysis evaluated a pool type design that relies on forced circulation of the primary coolant, a conventional steam power conversion system, and a passive decay heat removal system. The ATHENA computer code was used to simulate various transients without reactor scram, including a primary coolant pump trip, a station blackout, and a step reactivity insertion. The reactor design successfully met identified temperature limits for each of the transients analyzed.

1. Introduction

Heavy metal (lead or lead–bismuth) is currently being considered as a coolant for a variety of reactors (MacDonald and Buongiorno, 2001; Sekimoto et al., 2002; Spencer et al., 2000) designed to produce low-cost electricity and/or burn actinides. The economic advantages of heavy metal cooled reactors arise from the possibility of developing reactors with passive safety characteristics and long core lifetimes, combined with lower capital and operating costs than other fast reactors. The ability to burn actinides created by the current generation of light water reactors is also an attractive feature of these reactors.

The Idaho National Engineering and Environmental Laboratory (INEEL) and the Massachusetts Institute of Technology (MIT) are jointly assessing various design options for such reactors. The project goal is to identify and analyze key technical issues in core neutronics, thermal-hydraulics, fuels, materials and economics associated with the development of this reactor concept. MIT is responsible for optimizing the thermal-hydraulic design for steady-state power operation, while the INEEL is responsible for the transient analysis. The INEEL's principal analysis tool is the ATHENA computer code (Carlson et al., 1986), which was recently modified to represent lead–bismuth coolant (Davis and Shieh, 2000). ATHENA is incorporated as a compile time option in the RELAP5-3D computer code (RELAP5-3D Code Development Team, 2002), which has been used extensively for the thermal-hydraulic analysis of light water reactors. ATHENA retains all of the capabilities of RELAP5-3D, but allows the use of working fluids other than light water.

A thermal-hydraulic and economic evaluation of a preliminary design of the actinide-burner reactor that relied on natural circulation of lead–bismuth was completed recently (Davis et al., 2002). The design has since been modified to use forced convection of the primary coolant. This paper is concerned

* Tel.: +1-208-526-9470; fax: +1-208-526-0528.
E-mail address: cbd@inel.gov (C.B. Davis).

0029-5493/$ – see front matter
doi:10.1016/S0029-5493(03)00104-3

with a thermal-hydraulic evaluation of the modified actinide-burner reactor during several transients, including pump trip, station blackout, and reactivity insertion. The transients were simulated without reactor scram to demonstrate the safety margins inherent in the reactor design. The safety margins for the reactor were determined by comparing maximum calculated temperatures to the transient limits determined by MacDonald and Buongiorno (2001). The remainder of this paper describes the actinide-burner reactor, the ATHENA model of the actinide-burner reactor, calculated transient results, conclusions, and references.

2. Description of the actinide-burner reactor

The reactor primary coolant system and the reactor vessel auxiliary cooling system (RVACS), which passively removes core decay heat, are illustrated in Fig. 1. Fig. 2 provides a top view of the vessel showing the core and heat exchanger layout and defines the angles listed in Fig. 1. The primary circuit operates near atmospheric pressure and is contained within the reactor vessel, which is further contained within a guard vessel. This configuration results in an integrated, compact system. The annulus between the core barrel and the reactor vessel is used to accommodate the heat exchangers and the circulation pumps.

The reactor features a dual-free-level design. The vessel is separated into different regions by the core barrel, the reactor vessel liner, divider plates, and seal plate. The flow path starts in the lower plenum and extends up through the reactor core and the chimney, which are contained within the core barrel. The upper core barrel contains many large holes that direct the coolant into portions of the annulus between the core barrel and the reactor vessel liner. The upper plenum, which is defined as the region above the first level of holes in the core barrel, below the upper head, and inside the vessel liner, contains the free level of the hot pool. An inert cover gas fills the space between the hot free level and the upper head. The lead–bismuth coolant turns around within the upper plenum and flows downward through portions of the annulus between the core barrel and reactor vessel liner. This annular region contains four counter-flow heat exchangers. The lead–bismuth flows down on the shell side of the heat exchangers while the secondary coolant flows up on the tube side. After exiting the bottom of the heat exchangers, the coolant flows down through a downcomer region until reaching holes in the reactor vessel liner located near the elevation of the seal plate. These holes direct the coolant into the annular gap between the liner and the reactor vessel, which is called the vessel riser. The coolant flows upward through the riser until reaching holes located in the upper liner which direct the fluid into annular regions containing two reactor coolant pumps. The coolant flows down through these annular regions, which are referred to as the pump downcomer, until flowing through the coolant pumps and seal plate into the lower plenum. The upper portions of the vessel riser and the pump downcomer regions are connected to the cover gas in the upper plenum. These connections result in the formation of free levels of relatively cold fluid in the riser and pump downcomer. The levels of these cold pools are considerably lower than the level in the hot pool because of the pressure loss across the heat exchangers. The reactor vessel liner was placed so that the velocity of the lead–bismuth coolant in the vessel riser and pump downcomer during normal operation would not exceed the 2 m/s limit identified by MacDonald and Buongiorno (2001).

The reactor core contains 157 fuel bundles, each with 312 fuel rods. Each bundle contains gas filled streaming tubes located at the center and periphery of the bundle to promote neutron leakage and reduce the reactivity worth due to coolant voiding. The fuel contains a mixture of plutonium, minor actinides, thorium, uranium, and zirconium. The fuel is clad with a high-performance steel alloy. The gap between the fuel and the cladding is filled with liquid lead to reduce the operating temperature of the fuel.

The RVACS design is similar to that used in S-PRISM (Boardman et al., 2000). The RVACS consists of the reactor vessel, a guard vessel, a perforated plate, a collector cylinder, and a reactor silo. Atmospheric air is drawn into the gap between the guard vessel and collector cylinder to provide the ultimate heat sink. The design includes several features to enhance the decay heat removal. These features include using lead–bismuth in the gap between the reactor and guard vessels, a perforated plate in the gap between the guard vessel and the collector cylinder to increase the heat transfer area, and fins or boundary layer trips on the surface of the guard vessel and collector

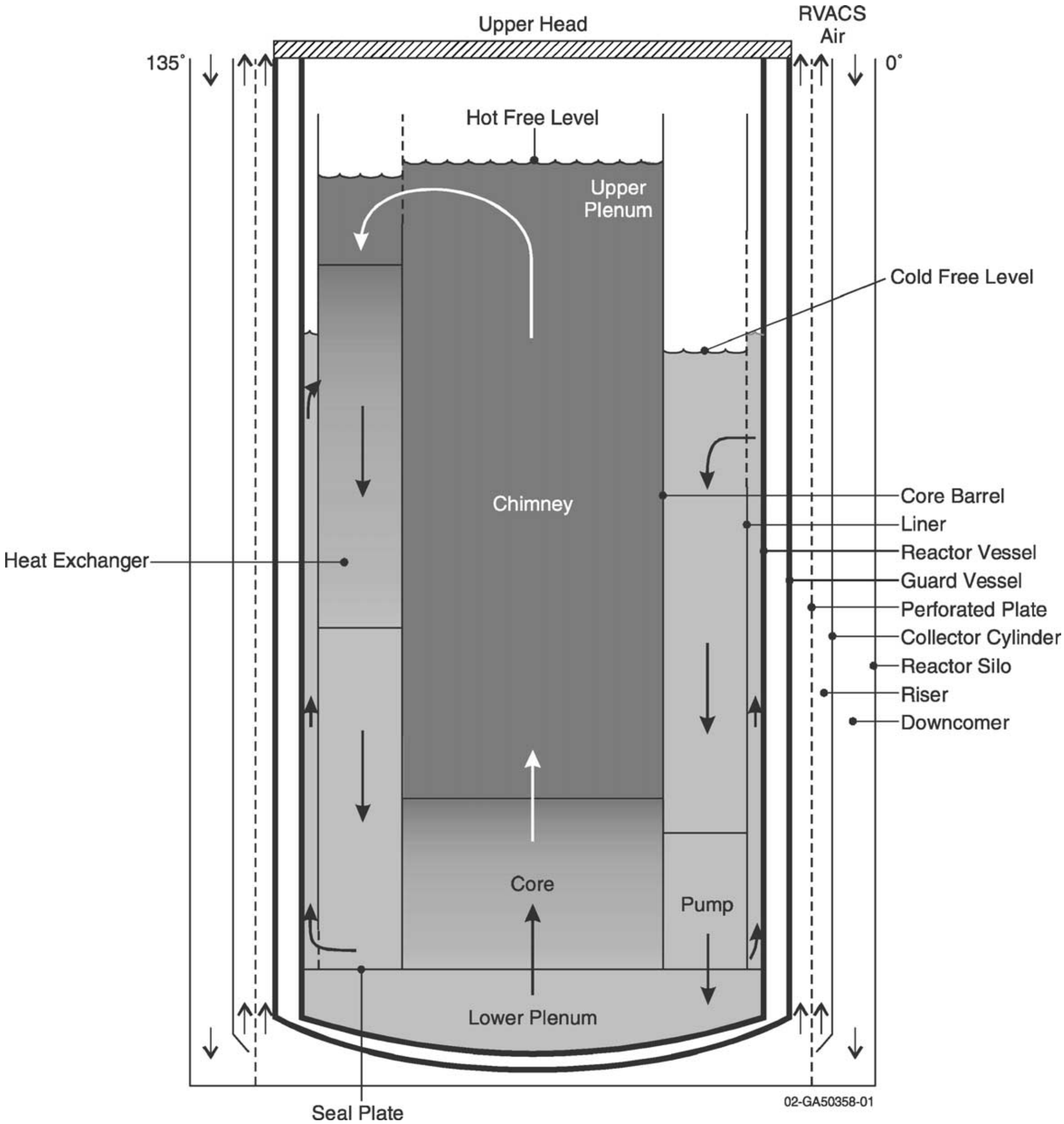

Fig. 1. Layout of the actinide-burner reactor.

cylinder. The gaps on either side of the perforated plate are large enough that the hydraulic resistance associated with the plate does not significantly limit the flow of air.

The actinide-burner reactor is designed with safety in mind. The lack of external loops and the use of a guard vessel essentially eliminate the possibility of loss-of-coolant accidents. The reactor vessel is sized so that the RVACS can passively and safely remove the decay heat from a 700-MWth (300-MWe) core. The core is designed with relatively low reactivity swings over its lifetime. This allows a relatively small control rod worth that limits the consequences of control rod withdrawal accidents. The dual-free-level design al-

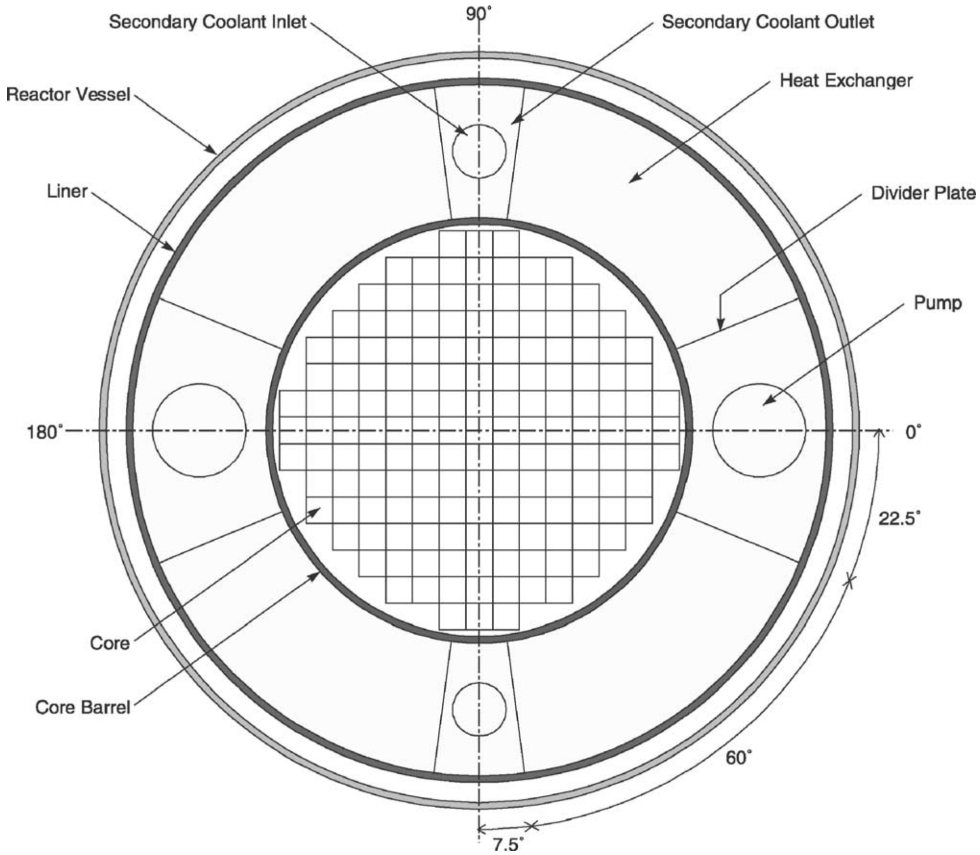

Fig. 2. Layout of the actinide-burner reactor—top view.

lows gas from a ruptured heat exchanger tube to separate at either the hot or cool levels before reaching the core. The design thus limits the positive reactivity insertion associated with limited voiding in the core following a tube rupture, regardless of the flow direction. The reactivity feedback due to complete voiding of the core is negative and the overall reactivity feedback is such that an inherent reactor shutdown occurs following loss-of-flow and loss-of-heat-sink events without scram. The inherent shutdown results in temperatures that meet material limits, thus assuring the integrity of the core and vessel.

3. ATHENA model description

The ATHENA model of the actinide-burner reactor is illustrated in Fig. 3, which shows the primary coolant system and the RVACS. The reactor core is represented by two parallel flow channels, one (Component 516) representing four high-powered fuel assemblies and the other (Component 510) representing the remaining 153 assemblies. The peaking factor of the four high-powered bundles is 1.16 and the maximum axial peaking factor is 1.155 (MacDonald and Buongiorno, 2001). The chimney and upper plenum are represented by Components 520, 530, 532, and 540. The downcomer, vessel riser, pump downcomers, and pumps are represented by Components 570, 580, 590, and 595, respectively.

The heat exchanger, Component 560, is based on MIT's optional design (Dostal et al., 2001) utilizing water as the secondary coolant. MIT's preferred design for the actinide-burner reactor currently utilizes carbon dioxide as the secondary coolant, but water is still being considered. The optional design utilizing

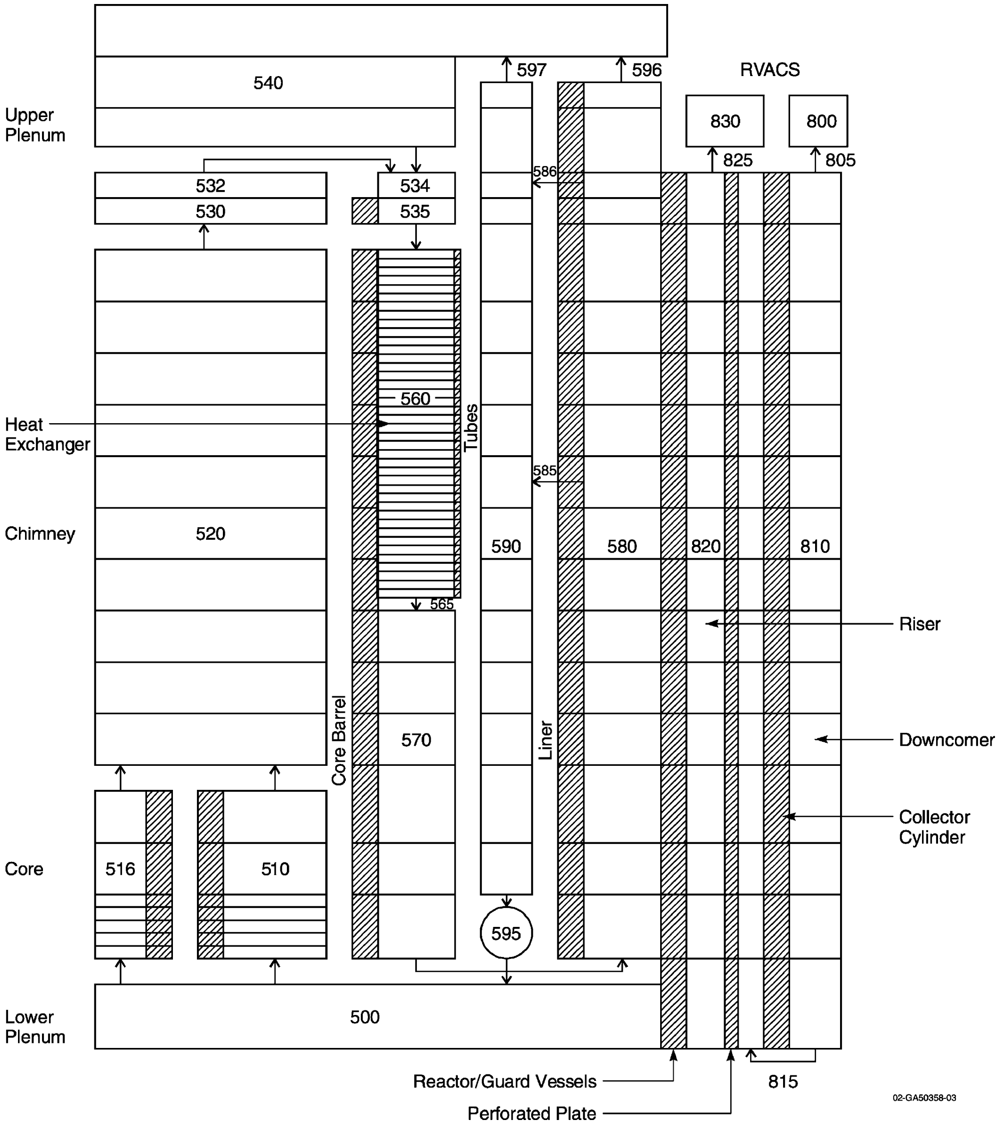

Fig. 3. ATHENA model of the primary coolant system and RVACS.

water as the secondary coolant was modeled in this analysis because carbon dioxide properties were not available in ATHENA when the calculations were performed. The potential effects of the secondary coolant on the transient results are discussed in Section 4. Calculations investigating the effect of the secondary coolant are planned. The ATHENA model represents a simplified secondary coolant system, in which feedwater flow is supplied with a time-dependent junction and the pressure is set with a time-dependent volume.

The average-powered and high-powered fuel rods, core barrel, reactor vessel liner, and heat exchanger tubes are represented with heat structures.

The RVACS is represented by Components 800–830 as shown in Fig. 3. The air supply and exhaust are represented with two time-dependent volumes, Components 800 and 830, that are set at atmospheric pressure. The RVACS downcomer and riser are represented by Components 810 and 820, respectively. The model represents all the major heat structures associated with the RVACS, including the reactor and guard vessels, the perforated plate, and the collector cylinder. The lead–bismuth in the gap between the reactor and guard vessels was modeled as a heat conducting material rather than as a fluid. The thermal conductivity of the lead–bismuth was increased to account for natural convection. Radiation enclosure models were used to represent the radiation heat transfer between the guard vessel, perforated plate, and the collector cylinder. The emissivity of these surfaces was set to 0.75, which is representative of the average measured value during the PRISM test program (Hunsbedt and Magee, 1988). The surface area of the perforated plate includes a 40% reduction to account for holes. The heat transfer coefficients at the surfaces of the guard vessel, perforated plate, and collector cylinder were doubled to account for the effects of fins or ribs. The factor of two increase is consistent with the heat transfer enhancement that can be achieved by the use of ribs to trip the boundary layer, such as used in S-PRISM (Boardman et al., 2000), due to the thermal entry effect. An insulating material is attached to the outside surface of the collector cylinder to protect the concrete silo from excessive temperature and to prevent heating of the air in the RVACS downcomer, which would reduce the air flow rate.

The model is believed to be conservative because the Dittus–Boelter correlation used by ATHENA predicts heat transfer coefficients that are, on average, about 20% lower than the lower bound of the data reported by Hunsbedt and Magee (1988) for an RVACS without a perforated plate and ribs. Furthermore, the RVACS heat transfer area is based on an active height for heat transfer that is reduced compared to that of the reactor vessel. For example, the primary coolant level during a station blackout is about 3% higher than the height assumed for the RVACS heat transfer. No credit is given for heat transfer from the cover gas. Although the heat transfer coefficient from the cover gas to the reactor vessel would be small, natural convection currents in the gap between the reactor and guard vessels would result in relatively effective heat transfer, at least up to the level of the lead–bismuth in that gap. The ATHENA RVACS model was previously benchmarked (Davis et al., 2002) against calculations performed by General Electric for the PRISM reactor, which did not contain liquid metal in the gap between the reactor and guard vessels, a perforated plate, or ribs. Although the current RVACS model is believed to be conservative, prototypical experiments would have to be conducted to validate the model prior to licensing the actinide-burner reactor.

The geometry of the actinide-burner reactor is summarized in Table 1.

The ATHENA model was used to obtain a steady state for the actinide-burner reactor at normal operating power. The results of the steady-state calculation are summarized in Table 2. The results of the steady-state calculations performed by MIT (Dostal et al., 2001) are presented for comparison where available. The feedwater flow rate used in ATHENA was 4.6% higher than the MIT value, causing a lower steam superheat. Preliminary calculations in which the MIT results were applied to the ATHENA steam generator model as boundary conditions yielded a total heat removal rate that was in reasonable agreement with, but 2.5% lower than, the MIT value. The feedwater flow was adjusted so that initial conditions predicted by ATHENA were consistent with the MIT value. As a result, the primary coolant temperatures calculated by ATHENA were in good agreement with those calculated by MIT. The maximum cladding temperature predicted by ATHENA was 2.1 °C below the 600 °C steady-state limit identified by MacDonald and Buongiorno (2001). The dual-free-level design results in relatively cold lead–bismuth contacting the reactor vessel wall during normal operation. The temperature of the guard vessel is lower than that of the reactor vessel because of cooling by the RVACS and easily meets the 430 °C limit for steady-state operation identified by MacDonald and Buongiorno.

Transient reactor power was calculated using a best-estimate point reactor kinetics model (MacDonald and Buongiorno, 2001). The model used a delayed neutron fraction of 0.0028 and prompt neutron generation lifetime of 2.71×10^{-5} s. The reactivity feedback model was used to simulate the effects of changes in fuel temperature, coolant temperature, and coolant density. The fuel temperature feedback coefficient was −0.0013 $/K, which represented the combined effects of Doppler (−0.0007 $/K) and fuel thermal expansion (−0.0006 $/K). The coolant temperature feedback coefficient, which represented the effects of radial expansion of the lower and upper core plates, was −0.0023 $/K. The coolant density feedback coefficient was −0.0003 $/kg/m^3, which

Table 1
Key parameters for the ATHENA model of the actinide-burner reactor

Parameter	Value
Core	
Fuel outer diameter (mm)	5.48
Cladding inner diameter (mm)	6.32
Cladding outer diameter (mm)	7.52
Fuel rod pitch-to-diameter ratio	1.3
Heated length (m)	1.3
Gas plenum length (m)	2.47
Number of fuel assemblies	157
Steam generator	
Lattice	Triangular
Number of tubes	13316
Tube pitch-to-diameter ratio	1.2
Tube inner diameter (mm)	13.9
Tube thickness (mm)	4.05
Tube length (m)	6.9
Miscellaneous	
Chimney length (m)	10.5
Core barrel inner diameter (m)	3.16
Core barrel thickness (m)	0.02
Vessel liner inner diameter (m)	5.26
Vessel liner thickness (m)	0.01
Gap between liner and reactor vessel (m)	0.16
Cover gas volume (m^3)	59
Pump diameter (m)	0.65
Reactor vessel	
Inner diameter (m)	5.60
Thickness (m)	0.05
Vessel length (m)	18.8
RVACS	
Gap between reactor and guard vessels (m)	0.05
Guard vessel thickness (m)	0.25
Gap between guard vessel and perforated plate (m)	0.197
Perforated plate thickness (m)	0.009525
Gap between perforated plate and collector cylinder (m)	0.1716
Collector cylinder thickness (m)	0.0254
Collector cylinder insulation thickness (m)	0.0508
Downcomer gap thickness (m)	0.5842
Active heat transfer length (m)	16.3

corresponds to a slightly positive coolant void worth, for small changes in density. Power-squared averaging was used to determine the weighting factors in the feedback model. The decay heat for the station blackout transient was conservatively based on 7.5 effective full-power years.

Table 2
Initial conditions for the actinide-burner reactor at rated power

Parameter	ATHENA	Dostal et al. (2001)
Primary coolant system		
Core power (MW)	700	700
Mass flow rate (kg/s)	54420	54420
Core inlet temperature (°C)	465.9	467
Core outlet temperature (°C)	554.3	554.9
Maximum cladding temperature (°C)	597.9	600
Maximum fuel temperature (°C)	701.2	682.8
Pumping power (MW)	3.8	3.1
Cover gas pressure (MPa)	0.1	NA[a]
Elevation of the hot pool level (m)	15.8[b]	NA
Elevation of the cold pool level (m)	11.8[b]	NA
Secondary coolant system		
Pressure (MPa)	15.0	15.0
Feedwater flow rate (kg/s)	333.3	316.7
Feedwater temperature (°C)	280	280
Steam superheat (°C)	178	206
RVACS		
Power removed (MW)	2.5	NA
Air flow rate (kg/s)	39.9	NA
Air inlet temperature (°C)	38.0	NA
Air exit temperature (°C)	100.1	NA
Maximum guard vessel temperature (°C)	363.2	NA

[a] Not applicable.
[b] Elevation is relative to the elevation of the seal plate, which is assumed to be 1.5 m above the bottom of the lower head of the reactor vessel.

4. Results

The ATHENA model was used to simulate various transients in the actinide-burner reactor, including a primary coolant pump trip, station blackout, step reactivity insertion, heat exchanger tube rupture, turbine stop valve closure, steam line break, and loss of feedwater preheating. In addition, a conceptual design of a reactor cleanup system was developed and the effects of loss of coolant from that system were evaluated. The transients were simulated without reactor scram to demonstrate the safety margins inherent in the reactor design. The safety margins were determined by comparing maximum calculated temperatures to the transient limits determined by MacDonald and Buongiorno (2001). These transient temperature limits were 725 °C for the fuel rod cladding, 1000 °C for the fuel, and 750 °C for the guard vessel. The temperature limit for the fuel cladding was based on fuel–clad-chemical-interaction considerations and is considered tentative pending additional material testing. The temperature limit for the fuel was based on restructuring considerations. The temperature limit for the guard vessel was based on mechanical considerations and may require its inner surface to be lined with a suitable corrosion-resistant material. Evaluations of the safety margins indicated that the most limiting transients were the primary coolant pump trip, station blackout, and 0.2 $ step reactivity insertion. Results from these limiting transients are presented below. MacDonald and Buongiorno (2002) present results for the other transients.

4.1. Primary coolant pump trip

The ATHENA model described previously was used to simulate a simultaneous trip of both primary coolant pumps combined with a failure to scram. The initial conditions for the calculation are presented in Table 2. The pump trip was initiated at 10 s. The feedwater flow was held constant until 60 s, when it was reduced to zero over a 10-s period. Continued feedwater flow would overfill the steam generators and result in excessive entrainment of liquid to the steam lines. The turbine header pressure was held constant.

The effect of the pump trip on the normalized coolant flow rate is shown in Fig. 4. The flow rate decreased smoothly following the pump trip at 10.0 s. Natural circulation was fully established at 86 s, when the head produced by the pumps became negative. The rate of the pump coastdown was controlled by the pump inertia, which was set to 4930 kg/m^2. With this inertia, the pump speed decreased to half of its initial value in 11.4 s. This rate of coastdown is similar to the 12-s value assumed by Sekimoto et al. (2002) in their analysis of a small, fast reactor cooled by lead–bismuth.

The effect of the pump trip on the fluid temperatures at the inlet and outlet of the core is shown in Fig. 5. The pump trip caused the core outlet temperature to increase rapidly. The cooling by the feedwater resulted in colder water reaching the core near 50 s. The feedwater was terminated 60 s after the pump trip. The termination of feedwater caused the core inlet temperature to begin to increase near 270 s. By the end of the calculation, the core inlet and outlet temperatures exceeded their initial values by about 100 °C on average. The reactivity feedback associated with the increase in primary coolant and fuel temperatures caused the core power to decrease as shown in Fig. 6. The reactor was essentially shutdown by 700 s, when the decay power exceeded the fission power.

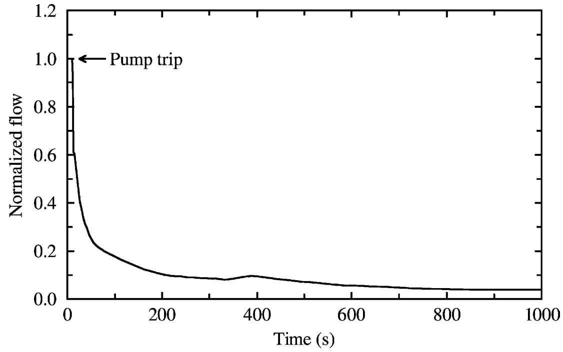

Fig. 4. Normalized coolant flow rate following the primary coolant pump trip.

The effect of the pump trip on the cladding temperature is also shown in Fig. 5. The temperature represents the axial maximum of the cladding temperatures in the high-powered channel. The cladding temperature increased following the pump trip until reaching a peak value of 709 °C at 58 s. The peak cladding temperature was below the transient temperature limit of 725 °C.

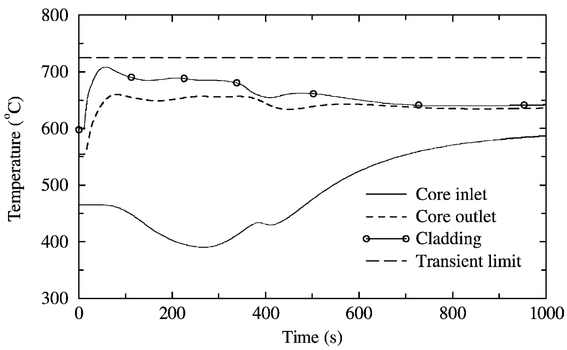

Fig. 5. Primary coolant and cladding temperatures following the primary coolant pump trip.

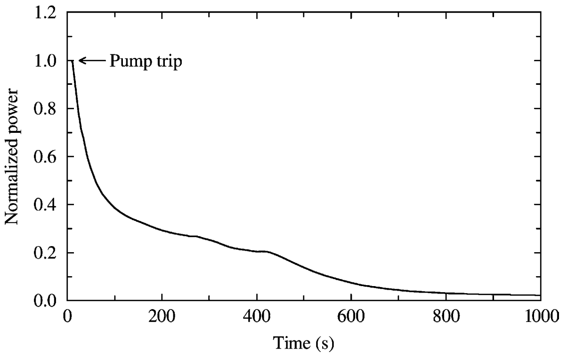

Fig. 6. Normalized core power following the primary coolant pump trip.

The reactivity feedback associated with cooling by the feedwater significantly affected the calculated results. For example, a sensitivity calculation showed that if the feedwater flow were not terminated, the cladding temperature would begin to increase at 160 s and would slightly exceed the transient limit of 725 °C after 430 s. The continued cooling by the feedwater reduced the fluid temperatures in the core compared to the values shown in Fig. 5. The associated reactivity feedback caused the reactor power to increase to nearly 80% of its initial value and resulted in a maximum cladding temperature of 732 °C. A second sensitivity calculation showed that terminating the feedwater at 140 s resulted in a second temperature peak that exceeded the first, but was still below the transient limit. These calculations indicate that there is ample time to terminate feedwater flow or shutdown the reactor via a redundant reactivity control system following a primary coolant pump trip without scram. The response of the actinide-burner reactor with carbon dioxide as the secondary coolant would probably be better than that described above for water. The secondary inlet temperature is more than 100 °C higher with carbon dioxide than with water (MacDonald and Buongiorno, 2001). Consequently, the temperature of the lead–bismuth exiting the heat exchanger would decrease less with carbon dioxide than with water, which would limit the reactivity feedback.

The relatively low coefficient of thermal expansion for lead–bismuth results in a modest pressure increase during this event. The maximum cover gas pressure was 0.12 MPa, representing a 20% increase from the initial value.

Loss-of-flow events do not cause serious consequences in the actinide-burner reactor. The design meets the identified temperature limits following the trip of the reactor coolant pumps combined with a failure to scram when the pumps coasts down to half of their initial speed in 11.4 s. The feedwater needs to be tripped within a few minutes of the initiating event or a redundant reactivity control system is required to provide long-term protection for the failure to scram.

4.2. Station blackout

The performance of the actinide-burner reactor during a station blackout was also evaluated using the ATHENA model described previously. The station blackout was assumed to cause a loss of main and auxiliary feedwater and a reactor pump trip. A failure to scram was also assumed. The initial conditions for the calculation are presented in Table 2. The accident was initiated at 0 s.

Fig. 7 shows the calculated maximum cladding temperature during the station blackout. Although it cannot be seen in the figure, an early temperature peak was calculated to occur at 60 s. This temperature peak was similar to that shown previously and was caused by the flow decreasing faster than the power following the pump trip. The maximum cladding temperature then decreased as reactivity feedback shutdown the reactor similar to that described previously. However, the core’s decay heat exceeded the power that was removed by the RVACS as shown in Fig. 8. The imbalance in heat generation and removal caused the

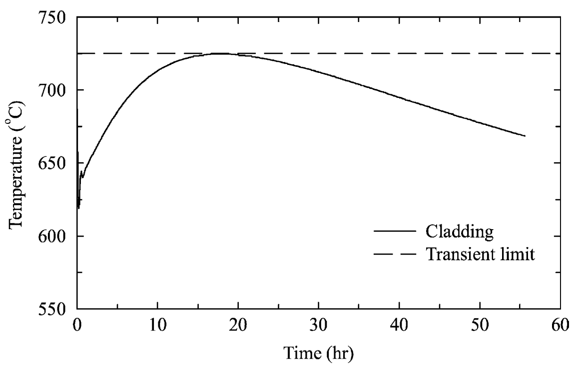

Fig. 7. Maximum cladding temperature during a station blackout.

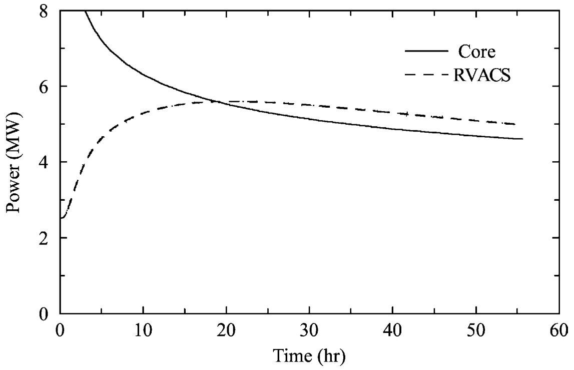

Fig. 8. A comparison of core decay power and RVACS heat removal during a station blackout.

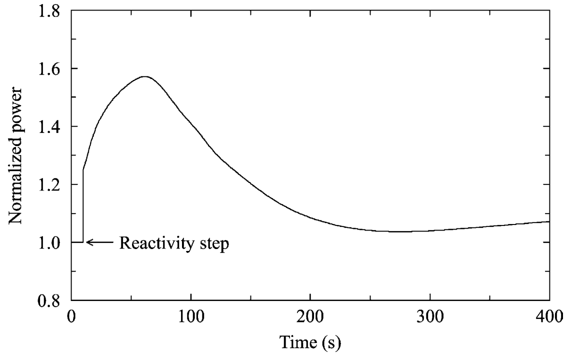

Fig. 9. Normalized core power following a 0.2 $ step reactivity insertion.

primary coolant system to heat up gradually until the cladding reached a maximum temperature near 18 h. After this time, the RVACS removed more power than the core generated and the temperature decreased slowly. The peak cladding temperature was 725 °C, equaling the transient limit.

The actinide-burner reactor just meets the identified cladding temperature limit following a station blackout with a failure to scram. Thus, the RVACS provides adequate long-term cooling for this accident. It is expected that the actual margin between the peak cladding temperature and the temperature limit is greater than that shown in Fig. 7 because of the conservative assumptions relative to the RVACS heat transfer performance described previously. Because the heat capacity of the secondary coolant is small compared to that of the primary, the calculated results for this transient are not expected to be sensitive to the choice of secondary coolant.

4.3. Step reactivity insertion

The performance of the actinide-burner reactor following the insertion of a 0.2 $ reactivity step was evaluated using the ATHENA model described previously. A failure to scram was assumed for this transient. The feedwater flow, turbine header pressure, and speed of the primary coolant pumps were assumed to remain constant during the transient. The 0.2 $ value represents the maximum worth of a single control rod. The maximum withdrawal speed of a control rod has not been determined, but is clearly bounded by a step insertion of reactivity. The initial conditions for the calculation are presented in Table 2. The accident was initiated at 10 s.

The withdrawal of a control rod could cause the power peaking near the affected fuel element to increase above the values assumed in the model. However, the point kinetics model cannot simulate the localized neutronic effects of the control rod withdrawal. Thus, this analysis will concentrate on the neutronic and thermal-hydraulic response of the entire core.

Fig. 9 shows the effect of the step reactivity insertion on the normalized reactor power. The reactor was operating at steady state prior to 10 s, when 0.2 $ of reactivity was instantly added to the core. The step reactivity insertion caused the reactor power to increase rapidly. The power increase caused negative reactivity feedback due to the increase in fuel and coolant temperature. The reactivity feedback increased near 60 s when hotter coolant first reached the inlet to the core (see Fig. 10) and the power reached its maximum value. The core power was nearly in equilibrium with the power removed by the heat exchangers when the calculation was terminated at 400 s.

The effects of the step insertion on the maximum cladding and primary coolant temperatures are shown in Fig. 10. The increasing temperature of the fluid at the core inlet caused the peak cladding and fuel temperatures to occur about 50 s after the peak power occurred. The peak cladding temperature remained below the transient limit.

The calculations indicated that the actinide-burner reactor meets the identified temperature limits follow-

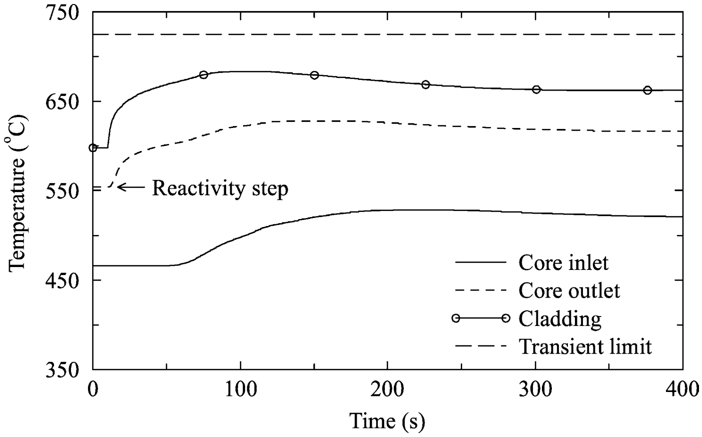

Fig. 10. Primary coolant and maximum cladding temperatures following a 0.2 $ step reactivity insertion.

Table 3
Summary of transient results

	Peak temperature (°C)		
	Cladding	Fuel	Guard vessel
Limit	725	1000	750
Transient			
Primary coolant pump trip without scram	709	753	382
Blackout without scram	725	759	532
0.2 $ step reactivity insertion without scram	684	828	369

ing a 0.2 $ step reactivity insertion. The point kinetics model indicates that a scram is not required to provide protection for this accident.

5. Conclusions

The results of the ATHENA calculations are summarized in Table 3. The table identifies the initiating event, the calculated peak temperatures for the cladding, fuel, and guard vessel, and the corresponding temperature limits identified by MacDonald and Buongiorno (2001).

Table 3 shows that the cladding temperature was always closer to its limit than the fuel and guard vessel temperatures. Thus, the cladding temperature is the most limiting parameter for the actinide-burner reactor. The station blackout is the most limiting transient. The station blackout coupled with a failure to scram produced a peak cladding temperature that was equal to the transient limit.

The station blackout without scram results in two cladding temperature peaks. The first temperature peak occurs about 1 min after the start of the event and is caused by the flow decreasing faster than the power following the trip of the primary coolant pumps. The pumps must coast down to half of their initial speed in 11.4 s or more in order to meet the cladding thermal limit. The second temperature peak occurs about 18 h after the start of the event and is associated with the balance between the power generated by the core and that removed by the RVACS.

The analysis shows that the actinide-burner reactor exhibits excellent safety characteristics. The reactor passively met the identified temperature limits for all the transients evaluated. Reactor scram was not required for any of the cases considered. Although the cladding temperature reached the thermal limit following the station blackout, this transient requires loss of main and auxiliary feedwater for an extended period of time coupled with a failure to scram and is very improbable. This event did not result in any fuel damage in the actinide-burner reactor whereas significant fuel damage could occur in a comparable event in operating light water reactors.

The current analysis provides a preliminary evaluation of the safety characteristics of the actinide-burner reactor. Detailed analyses simulating multi-dimensional hydraulic and neutronic effects are required to fully characterize the safety of the actinide-burner reactor.

Acknowledgements

This work was supported through the INEEL Long-Term Research Initiative Program under DOE Idaho Operations Office Contract DE-AC07-99ID13727.

References

Boardman, C.E., Dubberly, A.E., Carroll, D.G., Hui, M., Fanning, A.W., Kwant, W., 2000. A description of the S-PRISM plant, ICONE-8168. In: Proceedings of ICONE-8, Baltimore, MD, April 2–6.

Carlson, K.E., Roth, P.A., Ransom, V.H., 1986. ATHENA Code Manual, vol. I. Code Structure, System Models, and Solution Methods. Idaho National Engineering Laboratory, EGG-RTH-7397.

Davis, C.B., Shieh, A.S., 2000. Overview of the use of ATHENA for thermal-hydraulic analysis of systems with lead–bismuth coolant. In: Proceedings of ICONE-8, Baltimore, MD, April 2–6.

Davis, C.B., Kim, D., Todreas, N.E., Kazimi, M.S., 2002. Core power limits for a lead–bismuth natural circulation actinide burner reactor. Nucl. Eng. Des. 213, 165–182.

Dostal, V., Todreas, N.E., Hejzlar, P., Kazimi, M.S., 2001. Power Conversion Cycle Selection for the LBE Cooled Reactor with Forced Circulation. Massachusetts Institute of Technology, MIT-ANP-TR-084.

Hunsbedt, A., Magee, P.M., 1988. Design and performance of the PRISM natural convection decay heat removal system. In: Proceedings of the International Topical Meeting on Safety of Next Generation Power Reactors, Seattle, Washington, May 1–5.

MacDonald, P.E., Buongiorno, J., 2001. Design of an actinide burning, lead or lead–bismuth cooled reactor that produces low cost electricity. FY-01 Annual Report, Idaho National Engineering and Environmental Laboratory, INEEL/EXT-01-01376, MIT-ANP-PR-083.

MacDonald, P.E., Buongiorno, J., 2002. Design of an actinide burning, lead or lead–bismuth cooled reactor that produces low cost electricity. FY-02 Annual Report, Idaho National Engineering and Environmental Laboratory, INEEL/EXT-02-01249, MIT-ANP-PR-092.

RELAP5-3D Code Development Team, 2002. RELAP5-3D© Code Manual, vol. 1. Code Structure, System Models and Solution Methods, INEEL-EXT-98-00834 Revision 2.0.

Sekimoto, H., Makino, S., Nakamura, K., Kamishima, Y., Kawakita, T., 2002. Small LBE-cooled fast reactor for expanding market, ICONE-22049. In: Proceedings of ICONE-10, Arlington, Virginia, April 14–18.

Spencer, B.W., Hill, R.N., Wade, D.C., Hill, D.J., Sienicke, J.J., Khalil, H.S., Cahalan, J.E., Farmer, M.T., Maroni, V.A., Leibowitz, L., 2000. An advanced modular HLMC reactor concept featuring economy, safety, and proliferation resistance, ICONE-8145. In: Proceedings of ICONE-8, Baltimore, MD, April 2–6.